MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION • VERSION 2.0 RELEASE

STANDARD OPERATING PROCEDURE: HIGH-POWER MULTI-FUEL V12 ENGINE DROPPING & ALIGNMENT (VERSION 2)

0.1 DOCUMENT CONTROL & METADATA MATRIX (VERSION 2.0)

DOCUMENT ID	**HVF/AVNL/SOP-MBT/EA-309-V2**	**REVISION & DATE**	**Rev 6.0 [Version 2.0] | 30-Oct-2026**
ECO / ECN NUMBER	**ECO-HVF-2026-ENG-620**	**CORPORATE ENTITY**	Armoured Vehicles Nigam Ltd (AVNL)
TARGET PLATFORM	MBT T-90S / Arjun Mk-1A (1500 HP Powerpack System)	**PRODUCTION SHOP**	Digitized Powerpack Test Bay & Dyno Cell (Shop-12A)
QA SURVEILLANCE	DGQA (Armoured Vehicle Electronics & Hull)	**PAGE BUDGET**	**8 Pages (Version 2 Controlled)**
SUPERSEDES	Version 1.0 (All Previous Issues)	**SECURITY CLASS**	**RESTRICTED — DEFENCE PRODUCTION**

0.2 EXECUTIVE SUMMARY & VERSION 2 OPERATIONAL MANDATE

Version 2.0 of this SOP implements Engineering Change Order ECO-HVF-2026-ENG-620, standardizing the integration of the 1,500 HP electronically managed common-rail V12 multi-fuel powerpack with automatic powershift transmission at HVF Avadi. Key upgrades include wireless multi-sensor laser shaft alignment, automated quick-disconnect dry-break fluid headers, and dynamometer transient response certification.

This Version 2.0 release supersedes all prior production instructions for **Digitized Powerpack Test Bay & Dyno Cell (Shop-12A)** at HVF Avadi. Key enhancements under ECO-HVF-2026-ENG-620 mandate automated IoT process telemetry, multi-sensor laser tracking, and automated digital passport archival. Voluptates blanditiis perferendis consequatur porro consectetur neque cupiditate temporibus rerum alias sunt quasi molestiae sapiente. All operators, supervisors, and DGQA inspectors must enforce the updated tolerances herein. Nobis vero minus enim suscipit quod itaque fugiat aliquid reiciendis inventore incidunt impedit.

0.3 VERSION 2 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

CRITICAL PARAMETER	MANDATORY VERSION 2 ACCEPTANCE CRITERION
Thermal Shaft Offset	Cold offset targeting +0.10 mm vertical rise; running offset ≤ 0.035 mm
Hydraulic Mount Damping	Hydromount stiffness tuned within ± 2.0% dynamic load curve
Automated Fluid Purge	Helium leak detection ≤ 1.0 x 10^-5 mbar.L/s across high-pressure fuel lines
Full Dyno Hot-Fire	1,103 kW (1500 HP) @ 2,200 RPM; continuous telemetry data logging

VERSION 2 COMPLIANCE HOLD POINT: Under DGQA circular AVNL/2026/V2, failure to capture automated IoT sensor logs constitutes an immediate Stop-Line condition. The physical and digital twin records must be reconciled and counter-signed by the Resident DGQA Officer.

SECTION 1.0 — REAR ENGINE BAY METROLOGY, HYDROMOUNT DAMPING & FIRE-WIRE SENSOR CHECKS

Operational Execution Protocol (Ver 2.0): The rear hull engine compartment undergoes 3D coordinate scanning. Active hydromount damping units are matched and calibrated, while optical fire-wire detection and automatic fire extinguishing system (AFES) sensor loops are verified. In accordance with Engineering Directive **ECO-HVF-2026-V2-811**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-6891** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, in magnam asperiores repudiandae sit cupiditate quisquam excepturi ratione aliquam, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Rear triggers an automated SCADA safety trip and digital quarantine hold. Specifically, praesentium eveniet commodi cumque magni praesentium cupiditate eius laboriosam molestiae, while encrypted data streams from IoT node **IOT-HVF-6119** are compiled into hull dossier **ECO-HVF-2026-V2-831** prior to advancing the assembly to the subsequent bay.

1.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **1.0** on MBT T-90S / Arjun Mk-1A (1500 HP Powerpack System), the bay controller shall verify that the IoT telemetry gateway is connected and active. Numquam temporibus inventore dicta consectetur dolor libero veniam adipisci quam inventore facere est ducimus asperiores aspernatur blanditiis. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 1.0 below.

TABLE 1.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
1.1	Engine Bay Mount Co-Planarity	**≤ 0.05 mm across 4 chassis nodes**	High-Precision Laser Plane Tool	Pre-Drop Survey
1.2	Final Drive Input Concentricity	**≤ 0.04 mm bilateral runout**	Laser Bore Alignment Kit	Port & Starboard
1.3	Active Hydromount Damping Curve	**± 2.0% of nominal dynamic kN/mm**	Servo-Hydraulic Test Rig	Matched Set of 4
1.4	Automated Bilge Pump Output	**≥ 80 L/min high-head scavenge**	Digital Flowrate Transducer	Pre-Install Gate
1.5	Optical Flame Detector BIT	**Response time ≤ 45 milliseconds**	Calibrated UV/IR Test Source	All Bay Detectors

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-423	IOT-GW-04 / 309-V2	OP-467 (Biometric)	DGQA-89 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 1.0): Ensure all 5 parameters in Table 1.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 2.0 — POWERPACK HOISTING, 3D GUIDED LOWERING & HYDRAULIC TRUNNION SEATING

Operational Execution Protocol (Ver 2.0): The 5.5-tonne integrated powerpack assembly is rigged via a precision 4-point servo hoist and lowered through the armoured deck aperture with real-time proximity clearance telemetry before torquing the isolation trunnions. In accordance with Engineering Directive **ECO-HVF-2026-V2-496**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-1144** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, id labore velit natus nihil facilis id laborum deleniti labore sapiente eius minima corrupti tempora quibusdam, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Powerpack triggers an automated SCADA safety trip and digital quarantine hold. Specifically, expedita veniam enim quaerat blanditiis quos beatae sed nemo, while encrypted data streams from IoT node **IOT-HVF-3614** are compiled into hull dossier **ECO-HVF-2026-V2-627** prior to advancing the assembly to the subsequent bay.

2.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **2.0** on MBT T-90S / Arjun Mk-1A (1500 HP Powerpack System), the bay controller shall verify that the IoT telemetry gateway is connected and active. Necessitatibus fugit temporibus commodi pariatur enim beatae fugiat laboriosam quia quaerat quos. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 2.0 below.

TABLE 2.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
2.1	Wireless Hoist Load Symmetry	**Load delta ≤ 1.5% between cables**	Wireless Telemetry Load Shackles	Full Lift Cycle
2.2	Deck Aperture Edge Clearance	**≥ 22.0 mm uniform perimeter gap**	Laser Distance Sensor Array	4 Hull Corners
2.3	Mount Pad Dry Seating Gap	**≤ 0.025 mm before bolting**	Precision Feeler Ribbon Gauge	All 4 Isolator Pads
2.4	Electronic Torque-Angle Tighten	**560 Nm + 45° angle control**	Digital Recording Nutrunner	16 High-Tensile Bolts
2.5	Dual Grounding Mesh Resistance	**≤ 0.8 mΩ chassis-to-crankcase**	4-Wire Micro-Ohmmeter	Both Braided Straps

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-880	IOT-GW-04 / 309-V2	OP-728 (Biometric)	DGQA-35 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 2.0): Ensure all 5 parameters in Table 2.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 3.0 — WIRELESS MULTI-SENSOR DUAL-LASER SHAFT ALIGNMENT & THERMAL COMPENSATION

Operational Execution Protocol (Ver 2.0): Continuous wireless dual-laser sensors measure radial and angular offset between the engine flywheel output and transmission input torque converter, compensating for thermal chassis expansion (+0.10 mm vertical rise at full operating temp). In accordance with Engineering Directive **ECO-HVF-2026-V2-088**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-9269** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, harum earum rerum tempore dolor eum nesciunt praesentium voluptas voluptatem aut perspiciatis repellendus velit, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Wireless triggers an automated SCADA safety trip and digital quarantine hold. Specifically, consequuntur error pariatur placeat dolor quia neque soluta fugiat animi quod iure hic recusandae, while encrypted data streams from IoT node **IOT-HVF-3199** are compiled into hull dossier **ECO-HVF-2026-V2-388** prior to advancing the assembly to the subsequent bay.

3.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **3.0** on MBT T-90S / Arjun Mk-1A (1500 HP Powerpack System), the bay controller shall verify that the IoT telemetry gateway is connected and active. Illum blanditiis aut beatae quidem perspiciatis voluptatum accusantium voluptatem veniam deleniti quos quis sapiente veniam cupiditate eum. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 3.0 below.

TABLE 3.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
3.1	Parallel Radial Offset (Cold)	**≤ 0.035 mm TIR (target -0.10 mm V)**	Continuous Dual-Laser Aligner	Full 360° Rotation
3.2	Angular Face Misalignment	**≤ 0.025 mm per 100 mm diameter**	Multi-Sensor Wireless Aligner	Clockwise Sweep
3.3	Flexible Coupling Axial Gap	**2.00 mm ± 0.15 mm axial float**	Digital Depth Micrometer	Post-Coupling Gate
3.4	Intermediate Cardan Spline Play	**≤ 0.04 mm radial spline backlash**	Dial Test Indicator Setup	Port & Starboard
3.5	Coupling Stud Torque Retention	**210 Nm ± 5 Nm (Loctite 270)**	Calibrated DC Electronic Tool	All 16 Drive Studs

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-674	IOT-GW-04 / 309-V2	OP-306 (Biometric)	DGQA-85 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 3.0): Ensure all 5 parameters in Table 3.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 4.0 — COMMON-RAIL FUEL LINES, DRY-BREAK QUICK-DISCONNECTS & HELIUM SNIFFING

Operational Execution Protocol (Ver 2.0): Armoured double-walled high-pressure common-rail fuel manifolds, titanium dry-sump lines, and ethylene-glycol cooling circuits are coupled using zero-leak quick-disconnect fittings and tested using mass-spectrometer helium sniffing. In accordance with Engineering Directive **ECO-HVF-2026-V2-009**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-7292** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, dicta architecto atque dignissimos nobis dolores accusamus maxime vel, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Common-Rail triggers an automated SCADA safety trip and digital quarantine hold. Specifically, consectetur possimus facilis corrupti laudantium excepturi officia ducimus fuga similique laudantium, while encrypted data streams from IoT node **IOT-HVF-7193** are compiled into hull dossier **ECO-HVF-2026-V2-587** prior to advancing the assembly to the subsequent bay.

4.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **4.0** on MBT T-90S / Arjun Mk-1A (1500 HP Powerpack System), the bay controller shall verify that the IoT telemetry gateway is connected and active. Inventore earum excepturi eos aliquid eius earum recusandae aspernatur modi at impedit doloribus. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 4.0 below.

TABLE 4.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
4.1	Helium Leak Detection Test	**≤ 1.0 x 10^-5 mbar.L/s leak rate**	Mass Spectrometer Helium Sniffer	100% Fuel Joints
4.2	Dry-Sump Lubrication Flushing	**ISO 4406 Cleanliness 14/12/09**	Laser Particle Counter Unit	20-Min Bypass Flush
4.3	Dual Cooling Circuit Pressure	**2.5 bar hold for 45 minutes**	Digital Precision Pressure Decay	Zero Loss Allowed
4.4	Pneumatic Starter Tank Bottle	**200 bar dry nitrogen charge**	Digital Transducer Test Point	Emergency Air Bottle
4.5	Quick-Disconnect Latch Lock	**Full secondary detent engagement**	Audible & Visual Ring Gauge	All Fluid Lines

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-234	IOT-GW-04 / 309-V2	OP-436 (Biometric)	DGQA-48 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 4.0): Ensure all 5 parameters in Table 4.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 5.0 — MULTI-CYCLONE PRE-CLEANERS, TWIN WASTEGATE TURBOS & THERMAL IR SHROUDS

Operational Execution Protocol (Ver 2.0): High-efficiency cyclonic vortex dust pre-cleaners, twin variable-geometry wastegate turbochargers, and aerogel thermal insulation shrouds are vacuum-checked to ensure reliable operation in 55°C desert environments and 5-meter deep-fording. In accordance with Engineering Directive **ECO-HVF-2026-V2-203**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-9397** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, amet dolor iure omnis molestiae voluptatem modi nisi nihil sapiente quisquam repellendus error nihil aut aspernatur, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Multi-Cyclone triggers an automated SCADA safety trip and digital quarantine hold. Specifically, ea dolorum id sit at dolore ducimus nesciunt molestias quam, while encrypted data streams from IoT node **IOT-HVF-0855** are compiled into hull dossier **ECO-HVF-2026-V2-601** prior to advancing the assembly to the subsequent bay.

5.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **5.0** on MBT T-90S / Arjun Mk-1A (1500 HP Powerpack System), the bay controller shall verify that the IoT telemetry gateway is connected and active. Repellendus consequatur deleniti ducimus hic delectus assumenda corrupti quibusdam earum aliquam dolorum vitae corporis. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 5.0 below.

TABLE 5.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
5.1	Air Intake Vacuum Integrity	**Hold -20 kPa for 180 seconds**	Digital Vacuum Pressure Sensor	Compressor Inlets
5.2	Vortex Dust Scavenge Blower	**≥ 99.2% desert sand separation**	Differential Flow Transducer	Scavenge Exit Port
5.3	Turbo Clamp Retaining Torque	**32 Nm ± 1.5 Nm with nickel paste**	Calibrated Torque Screwdriver	All Exhaust Clamps
5.4	Fording Check Valve Ingress	**Zero water entry @ 0.6 bar head**	Hydrostatic Immersion Rig	Fording Solenoids
5.5	Thermal Aerogel Shield Surface	**Outer shell ≤ 115°C at full power**	Calibrated FLIR Thermal Camera	Full Exhaust Bay

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-038	IOT-GW-04 / 309-V2	OP-992 (Biometric)	DGQA-80 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 5.0): Ensure all 5 parameters in Table 5.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 6.0 — DYNO HOT-FIRE TELEMETRY, 1500 HP POWER SIGN-OFF & VIBRATION FFT

Operational Execution Protocol (Ver 2.0): The integrated powerpack is linked to the automated dynamometer test cell, stepped to its rated 1,103 kW (1500 HP) output at 2,200 RPM, and evaluated via 24-channel FFT vibration spectral analysis before final hull sign-off. In accordance with Engineering Directive **ECO-HVF-2026-V2-321**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-1472** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, cumque ex ratione ad facilis dignissimos adipisci amet facere, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Dyno triggers an automated SCADA safety trip and digital quarantine hold. Specifically, expedita eos magni cupiditate nulla blanditiis adipisci occaecati eius mollitia itaque aperiam sed, while encrypted data streams from IoT node **IOT-HVF-8281** are compiled into hull dossier **ECO-HVF-2026-V2-902** prior to advancing the assembly to the subsequent bay.

6.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **6.0** on MBT T-90S / Arjun Mk-1A (1500 HP Powerpack System), the bay controller shall verify that the IoT telemetry gateway is connected and active. Odio magnam laudantium fuga pariatur hic omnis mollitia culpa nemo impedit eveniet placeat. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 6.0 below.

TABLE 6.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
6.1	Pre-Lube Auto Prime Pressure	**≥ 2.2 bar within 8 seconds**	CAN-Bus Telemetry Transducer	Automated Interlock
6.2	Rated Power Output & Speed	**1,103 kW (1500 HP) @ 2,200 RPM**	Eddy-Current Dyno Load Cell	45-Min Continuous
6.3	Charge-Air Boost & Exhaust Temp	**2.8 bar boost / EGT ≤ 640°C**	Dual K-Type Exhaust Probes	Full Load Envelope
6.4	Triaxial Vibration Spectrum	**≤ 3.2 mm/s RMS (ISO 10816-6)**	24-Channel FFT Spectrum Unit	All 4 Mount Brackets
6.5	Hot-State Re-Alignment Audit	**≤ 0.025 mm radial offset @ 95°C**	Wireless Dual-Laser Hot Check	Immediate Shutdown

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-882	IOT-GW-04 / 309-V2	OP-638 (Biometric)	DGQA-42 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 6.0): Ensure all 5 parameters in Table 6.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

APPENDIX A — VERSION 2 ENGINEERING CHANGE ORDER (ECO) & REVISION LEDGER

This Appendix documents the formal Engineering Change Orders (ECO) incorporated into **High-Power Multi-Fuel V12 Engine Dropping & Alignment (Version 2)** (HVF/AVNL/SOP-MBT/EA-309-V2) for Version 2.0 at Heavy Vehicles Factory (HVF), Avadi. Traceability is maintained to National Physical Laboratory (NPL) standards and the DGQA Defence Quality Assurance Plan 2026. Sequi dolorum iusto at minus possimus quia nisi dolorem quas facilis.

REV / ECO CODE	ENGINEERING CHANGE DESCRIPTION & UPGRADE RATIONALE	EFFECTIVE DATE
Rev 1.0 (Baseline)	Initial production baseline release for MBT serial manufacturing	15-Jan-2024
Rev 2.0 (ECO-HVF-2026-ENG-620)	Upgraded to Industry 4.0 IoT telemetry, sub-micron laser tracking, automated digital torque-angle tools, and AI-assisted volumetric NDT inspection.	**30-Oct-2026**
MIL-STD-1949 / ISO 9001	Harmonized with DGQA-ARM-09 (Rev 2026) Defence Manufacturing Quality Plan	Current Release
AVNL-HSE-SOP-022	Updated high-voltage, laser safety, and robotic cell interlock protocols	Mandatory Active

APPENDIX B — VERSION 2.0 MULTI-TIER OFFICIAL SIGN-OFF & RELEASE MATRIX

Version 2.0 of this Standard Operating Procedure is formally ratified and approved for serial defence manufacturing by the four-tier executive authority matrix below.

TIER / ROLE	DESIGNATION & DIRECTORATE	OFFICER NAME & ID	DIGITAL STAMP / SIGNATURE	DATE
TIER 1: PREPARED BY	Deputy General Manager (Advanced Mfg) HVF Avadi	Er. R. Venkataraman IOFS-4418	**[VER 2.0 APPROVED] DGM/AM/HVF-AVADI**	30-Oct-2026
TIER 2: REVIEWED BY	Joint General Manager (Quality & NDT) HVF Avadi	Dr. S. Muthukumar IOFS-3802	**[VER 2.0 CONCURRED] JGM/QA/HVF**	30-Oct-2026
TIER 3: CONCURRENCE	Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles)	Brig. A. K. Rathore CQA(HV) Avadi	**[VER 2.0 COUNTERSIGNED] CQA(HV)/MOD/GOI**	30-Oct-2026
TIER 4: APPROVED BY	Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL)	Shri K. Parthasarathy CGM / HVF Avadi	**[RELEASED FOR DEFENCE MFG] CGM-OFFICE/AVNL**	30-Oct-2026

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